

Approximate Stationary Analysis and Empirical Verification of Noise-Resilient Cooperation in the Iterated Prisoner's Dilemma

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Abstract: The Iterated Prisoner's Dilemma (IPD) is a canonical framework for studying the emergence and stability of cooperation in repeated strategic interactions. Although Tit-for-Tat (TFT) has historically been viewed as a robust cooperative strategy, prior research demonstrates that its performance deteriorates under implementation noise due to persistent retaliation cycles and increased susceptibility to mutual defection traps. This study introduces NAFT (Noisy Adaptive Forgiving Tit-for-Tat), a memory-1 strategy that incorporates controlled stochastic forgiveness to improve cooperative stability in noisy environments.

Using a Markov chain formulation, we derive first-order stationary approximations for TFT and NAFT under execution noise and analytically characterize their long-run state occupancies and expected payoffs. The analysis demonstrates that NAFT reduces conflict-state occupancy and conditionally improves long-run efficiency in noise regimes where mutual defection does not become a dominant absorbing state. To validate the theoretical results, extensive Monte Carlo simulations are conducted across a wide spectrum of noise levels to empirically estimate stationary distributions, mixing behavior, and payoff outcomes.

Simulation results show that NAFT maintains higher cooperative mass across most noise regimes and achieves payoff advantages over TFT in low-to-moderate noise conditions, with reversals observed at higher noise levels. Additionally, empirical findings reveal that TFT is substantially more sensitive to stochastic perturbations than predicted by simplified analytical approximations, frequently converging to persistent defection states. This observation reframes classical TFT optimality as a theoretical benchmark rather than a robust outcome in realistic noisy environments.

This study demonstrates that stochastic forgiveness can act as a mathematically grounded mechanism for stabilizing cooperation in noisy environments. These results contribute to evolutionary game theory, multi-agent coordination, and cooperative artificial intelligence by providing analytic insight into the design of noise-resilient cooperative strategies.

Keywords: Iterated Prisoner's Dilemma, Cooperation under Noise, Memory-1 Strategies, Markov Chains, Stationary Distributions, Tit-for-Tat, Forgiveness in Games, Evolutionary Game Theory, Multi-Agent Systems, Cooperative AI

● Introduction

- 1.1 The Iterated Prisoner's Dilemma (IPD) is a foundational framework for analyzing cooperation, reciprocity, and strategic stability in repeated interactions. Since Axelrod's seminal tournaments identified Tit-for-Tat (TFT) as a highly effective cooperative strategy, reciprocal policies have been widely studied as mechanisms that sustain cooperation across evolutionary, economic, and artificial intelligence contexts (4). TFT's defining simplicity—cooperating initially and thereafter mirroring the opponent's previous action—has established it as a canonical baseline for evaluating cooperative behavior. This work does not claim evolutionary stability or optimality, but focuses on stationary robustness under noise
- 1.2 Despite its theoretical appeal, subsequent research demonstrates that TFT is highly sensitive to implementation noise and perception errors. Even infrequent accidental defections can trigger extended retaliation cycles, reducing long-run cooperation and leading to persistent mutual defection (7; 8). These findings reveal a structural limitation of strict reciprocity: in stochastic environments, deterministic retaliation can destabilize

cooperative equilibria by trapping interactions in low-payoff states. This limitation has motivated the development of strategies that incorporate forgiveness or probabilistic recovery mechanisms to improve resilience under noise (11).

- 1.3 Parallel advances in evolutionary game theory have focused on memory-1 strategies, which condition behavior solely on the previous round's outcome. These strategies allow repeated interactions to be modeled as finite Markov chains, enabling analytical characterization of stationary distributions and long-run payoffs (16). While this framework provides powerful tools for evaluating strategic stability, much of the existing literature emphasizes tournament performance or evolutionary simulations rather than deriving explicit closed-form stationary solutions for noisy reciprocal strategies. Moreover, the relationship between stochastic forgiveness and stationary-state dynamics remains insufficiently formalized.
- 1.4 This study addresses these gaps by introducing *NAFT* (Noisy Adaptive Forgiving Tit-for-Tat), a memory-1 strategy that retains reciprocal behavior while incorporating controlled stochastic forgiveness. The primary contribution of this work is threefold. First, we derive first-order closed-form stationary approximations for TFT and NAFT under implementation noise, providing analytical expressions for conflict-state occupancy and cooperative mass. Second, we establish conditions under which stochastic forgiveness reduces conflict persistence and improves long-run efficiency relative to strict reciprocity. Third, we empirically validate the theoretical predictions through large-scale Monte Carlo simulations, examining convergence behavior, payoff outcomes, and noise sensitivity. Additionally, this work evaluates evolutionary robustness of NAFT within heterogeneous multi-strategy populations, extending analysis beyond pairwise interaction models.
- 1.5 Beyond theoretical game dynamics, these results contribute to the design of cooperative policies in multi-agent reinforcement learning and cooperative artificial intelligence, where agents must maintain coordination under uncertainty and execution errors (24; 25). By demonstrating how stochastic forgiveness reshapes stationary Markov dynamics, this study provides analytical insight into the structural mechanisms required to sustain cooperation in noisy strategic environments.

● Research Questions

- 2.1 This study investigates how stochastic forgiveness alters long-run cooperation dynamics in noisy Iterated Prisoner's Dilemma environments. Specifically, it addresses the following research questions:
 - RQ1.** How does implementation noise alter the stationary-state distribution and long-run cooperation levels of Tit-for-Tat (TFT) in repeated Prisoner's Dilemma interactions?
 - RQ2.** Under what conditions can a memory-1 strategy incorporating stochastic forgiveness (NAFT) achieve higher stationary cooperation mass (π_{CC}) and reduced conflict-state persistence relative to TFT under noise?
 - RQ3.** Can first-order stationary approximations analytically characterize the payoff performance of TFT and NAFT, and identify regimes in which stochastic forgiveness improves long-run efficiency?
 - RQ4.** To what extent do Monte Carlo simulations validate the closed-form stationary predictions for TFT and NAFT across varying noise levels?
 - RQ5.** How does stochastic forgiveness influence transition dynamics between cooperative, retaliatory (CD/DC), and mutual defection (DD) states in noisy repeated games?

● Literature Review

Reciprocity and Tit-for-Tat as a Canonical Benchmark

- 3.1 The Iterated Prisoner's Dilemma (IPD) has long served as a foundational framework for studying cooperation and strategic stability in repeated interactions. Axelrod's computational tournaments established Tit-for-Tat (TFT) as a highly effective reciprocal strategy, demonstrating that simple contingent cooperation can sustain mutually beneficial outcomes (4). TFT's deterministic rule—cooperate initially and thereafter replicate the opponent's previous action—has since become a standard baseline for evaluating cooperative strategies across evolutionary game theory and multi-agent systems.

- 3.2** Despite its historical prominence, TFT's long-run cooperative stability is typically evaluated under idealized deterministic assumptions. Understanding how TFT's stationary-state behavior changes under stochastic execution environments therefore remains essential for assessing its practical robustness. This motivates investigation of how implementation noise alters TFT's long-run state occupancy and cooperation stability (RQ1).

Noise Sensitivity and the Limits of Strict Reciprocity

- 3.3** Subsequent research demonstrated that TFT's performance deteriorates significantly under implementation noise and perception errors. Nowak and Sigmund (1992) showed that accidental defections can trigger persistent retaliation cycles, reducing long-run cooperation levels (7). Wu and Axelrod (1995) further quantified TFT's sensitivity, showing that even low noise probabilities can destabilize cooperative equilibria (8).
- 3.4** These findings highlight that strict reciprocity may be structurally vulnerable to stochastic disturbances. However, existing studies largely rely on evolutionary simulations or tournament outcomes, leaving the stationary-state mechanisms underlying retaliation persistence insufficiently formalized. This limitation motivates analysis of transition dynamics and defection trapping in noisy reciprocal strategies (RQ5).

Forgiveness-Based Strategies and Generous Reciprocity

- 3.5** Forgiveness mechanisms have been proposed to mitigate noise-induced retaliation cycles. Generous Tit-for-Tat (GTFT) introduces probabilistic cooperation following opponent defection, improving resilience to stochastic errors (Nowak & Sigmund, 1992; Wu & Axelrod, 1995). Such strategies demonstrate improved cooperative performance in evolutionary and tournament-based environments.
- 3.6** However, forgiveness-based strategies are commonly evaluated empirically rather than through closed-form stationary-state analysis. Consequently, the structural relationship between forgiveness parameters, stationary cooperation levels, and long-run payoff efficiency remains incompletely characterized. This motivates investigation of whether probabilistic forgiveness can systematically increase stationary cooperation mass and payoff performance relative to strict reciprocity (RQ2 and RQ3).

Markov Chain Analysis and Memory-1 Strategy Theory

- 3.7** Memory-1 strategies, which condition behavior only on the previous round outcome, provide an analytically tractable representation of repeated-game dynamics. These strategies induce finite Markov chains, allowing stationary distributions and expected payoffs to be derived explicitly (18). Press and Dyson (2012) extended this framework through Zero-Determinant strategies, demonstrating that players can control payoff relationships through stationary-state manipulation (16). Subsequent work showed that evolutionary success is closely linked to stationary-state occupancy patterns rather than short-term tactical performance (19; 21).
- 3.8** Although this theoretical framework enables analytic characterization of repeated-game equilibria, comparatively few studies apply closed-form stationary analysis to concrete reciprocal strategies under noise. This motivates deriving explicit stationary payoff expressions and identifying efficiency regimes analytically (RQ3).

Empirical Validation of Analytical Stationary Models

- 3.9** While analytical stationary-state models provide theoretical predictions of long-run behavior, empirical verification remains necessary to confirm convergence properties and robustness under stochastic simulation environments. Monte Carlo methods have been widely used to evaluate evolutionary and repeated-game dynamics, but direct empirical validation of closed-form stationary distributions for forgiving reciprocal strategies remains limited. This motivates systematic simulation-based validation across varying noise regimes (RQ4).

Research Gap and Positioning of the Present Study

- 3.10** The literature reveals three primary unresolved challenges. First, the stationary-state response of classical TFT under implementation noise remains incompletely characterized analytically. Second, formal comparisons between strict and forgiving reciprocal strategies using closed-form stationary payoff analysis remain scarce.

Third, the dynamic mechanisms through which stochastic forgiveness alters transition behavior and prevents persistent defection trapping remain insufficiently formalized.

- 3.11** This study addresses these gaps by introducing NAFT (Noisy Adaptive Forgiving Tit-for-Tat), a memory-1 strategy that incorporates controlled stochastic forgiveness. The proposed framework enables analytical derivation of stationary distributions, payoff comparisons, and transition dynamics, and empirically validates theoretical predictions through large-scale simulation.

● Theoretical Derivations

Markov Chain Formulation

- 4.1** Let the outcome space be:

$$\mathcal{S} = \{CC, CD, DC, DD\}$$

where the first letter denotes the focal player's action and the second denotes the opponent's. Each memory-1 strategy is parameterized by a vector $\mathbf{p} = (p_{CC}, p_{CD}, p_{DC}, p_{DD})$, where p_{XY} is the probability of cooperating after state XY .

- 4.2** Let $\epsilon \in (0, 1)$ denote action noise, meaning each intended action flips with probability ϵ . The interaction induces a Markov chain with transition matrix M and stationary distribution π satisfying:

$$\pi M = \pi, \quad \sum_{s \in \mathcal{S}} \pi_s = 1$$

Transition Dynamics for Tit-for-Tat (TFT)

- 4.3** Tit-for-Tat is defined as $\mathbf{p}_{TFT} = (1, 0, 1, 0)$. TFT cooperates if and only if the opponent cooperated in the previous round. Under noise, the execution probability $\hat{p}$ for an intended action p is:

$$\hat{p} = p(1 - \epsilon) + (1 - p)\epsilon$$

The key observation is that noise introduces transitions from CC into the retaliation loop $CD \leftrightarrow DC$.

Conflict Cycle Probability Mass Under TFT

- 4.4** Once an erroneous defection occurs from CC , TFT deterministically retaliates, producing a two-cycle: $CD \rightarrow DC \rightarrow CD \rightarrow \dots$. Let π_{CD}^{TFT} and π_{DC}^{TFT} denote the stationary probabilities. The flow balance condition implies $\pi_{CD}^{TFT} = \pi_{DC}^{TFT}$.

- 4.5** Noise injects probability mass into the conflict cycle at rate ϵ and exits only when another noise event restores CC . Balancing the dominant first-order probability flows yields an approximate expression for the conflict mass and subsequent cooperative mass:

$$\pi_{CD}^{TFT} + \pi_{DC}^{TFT} = \frac{2\epsilon}{1 + 2\epsilon} \implies \pi_{CC}^{TFT} = \frac{1}{1 + 2\epsilon}$$

This expression is a first-order low-noise approximation obtained by neglecting higher-order occupation of the mutual defection state.

Expected Payoff of TFT

- 4.6** The expected payoff per round is defined as:

$$\Pi_{TFT} = \pi_{CC}^{TFT} R + \pi_{CD}^{TFT} S + \pi_{DC}^{TFT} T + \pi_{DD}^{TFT} P$$

Assuming $\pi_{DD}^{TFT} \approx 0$ and using the symmetry $\pi_{CD}^{TFT} = \pi_{DC}^{TFT}$, we obtain:

$$\Pi_{TFT} = R \left(\frac{1}{1 + 2\epsilon} \right) + \frac{S + T}{2} \left(\frac{2\epsilon}{1 + 2\epsilon} \right)$$

Transition Dynamics for NAFT

- 4.7** NAFT is defined as $\mathbf{p}_{NAFT} = (1, \alpha, 1, \beta)$ with $0 < \beta < \alpha < 1$. Unlike TFT, NAFT forgives isolated defections with probability α , allowing the transition $CD \rightarrow CC$ with probability $\alpha(1 - \epsilon)$. This introduces an escape route from the retaliation cycle.

Closed-Form Conflict Mass Under NAFT

- 4.8** Let $\pi_{conf}^{NAFT} = \pi_{CD}^{NAFT} + \pi_{DC}^{NAFT}$. Solving the modified flow equations yields:

$$\pi_{conf}^{NAFT} = \frac{2\epsilon}{\alpha + 2\epsilon}$$

Since $0 < \alpha < 1$, it follows that $\pi_{conf}^{NAFT} < \pi_{conf}^{TFT}$, and the cooperative mass is:

$$\pi_{CC}^{NAFT} = 1 - \frac{2\epsilon}{\alpha + 2\epsilon}$$

Expected Payoff of NAFT

- 4.9** The long-run payoff for NAFT is given by:

$$\Pi_{NAFT} = R \left(1 - \frac{2\epsilon}{\alpha + 2\epsilon} \right) + \frac{S + T}{2} \left(\frac{2\epsilon}{\alpha + 2\epsilon} \right)$$

Analytical Payoff Difference and Conditional Dominance

- 4.10** To evaluate the relative efficiency of the proposed strategy, we define the expected payoff difference as $\Delta = \Pi_{NAFT} - \Pi_{TFT}$. By substituting the closed-form stationary distributions derived in Eq. (4.4) and Eq. (4.7), and maintaining the primary modeling assumption that the system remains within a cooperative regime (where mass in the mutual defection state π_{DD} is negligible), we obtain:

4.11

$$\Delta = \left(R - \frac{T + S}{2} \right) \left(\frac{2\epsilon}{1 + 2\epsilon} - \frac{2\epsilon}{\alpha + 2\epsilon} \right) \quad (1)$$

- 4.12** Given the standard Prisoner's Dilemma constraints where $R > \frac{T+S}{2}$ and our definition of the forgiveness parameter $0 < \alpha < 1$, it follows that:

4.13

$$\frac{2\epsilon}{\alpha + 2\epsilon} < \frac{2\epsilon}{1 + 2\epsilon} \quad (2)$$

- 4.14** This inequality indicates, under the stated low-noise approximation, that NAFT maintains a lower occupancy in conflict states (CD and DC) compared to TFT. This leads to the following proposition:

Assumption 1 (Cooperative Regime Approximation). For low-to-moderate noise levels, the stationary probability of the mutual defection state remains small relative to cooperative and conflict states. Analytical payoff expressions therefore approximate the stationary distribution by neglecting higher-order π_{DD} contributions. The validity of this approximation is evaluated empirically in Section 5.

Proposition 1 (Conditional Payoff Improvement). Under Assumption 1 and for sufficiently small execution noise ϵ , NAFT yields higher expected payoff than TFT for any forgiveness parameter $0 < \alpha < 1$.

It is important to qualify that this efficiency advantage is *conditional*. As noise ϵ increases, the assumption that $\pi_{DD} \approx 0$ becomes harder to maintain for strict reciprocity strategies like TFT. As will be demonstrated in the empirical results (Table 2), when the system enters the “DD Trap,” the analytic efficiency advantage derived here serves as a theoretical upper-bound for performance rather than a definitive predictor of the long-run equilibrium.

● Methodology

Experimental Objective

- 5.1** This study evaluates the robustness of Tit-for-Tat (TFT) and Nosi Adaptive Forgiving Tit-for-Tat (NAFT) in the Iterated Prisoner's Dilemma (IPD) under stochastic execution errors. The primary objective is to compare stationary-state dynamics, payoff stability, and noise resilience using both closed-form theoretical analysis and large-scale Monte Carlo simulation.
- 5.2** The investigation focuses on intrinsic stability properties under symmetric self-play, where both agents employ identical strategies. The study does not attempt to evaluate evolutionary invasion dynamics, but rather examines stationary cooperation persistence under repeated interaction.

Game Model and Payoff Structure

- 5.3** The IPD is modeled using the standard Prisoner's Dilemma payoff ordering:

5.4

$$T > R > P > S, \quad 2R > T + S \quad (3)$$

- 5.5** Where:

- R = Reward for mutual cooperation
- T = Temptation payoff for unilateral defection
- P = Punishment for mutual defection
- S = Sucker's payoff

Unless otherwise stated, simulations employ the canonical payoff set:

$$(T, R, P, S) = (5, 3, 1, 0)$$

This normalization preserves the ordinal Prisoner's Dilemma structure while enabling direct numerical comparison. Analytical results remain invariant under affine payoff transformations.

Each interaction is assumed to be infinitely repeated, allowing convergence toward stationary Markov equilibria. In simulations, infinite repetition is approximated by long finite horizons ($T = 1000$), which are sufficient for convergence to stationary behavior.

Strategy Definitions

Tit-for-Tat (TFT)

- 5.6** TFT cooperates in the initial round and subsequently replicates the opponent's previous action. TFT is known to sustain cooperation in deterministic environments but is susceptible to entering oscillatory retaliation cycles ($CD \leftrightarrow DC$) when execution errors occur.

Nosi Adaptive Forgiving Tit-for-Tat (NAFT)

- 5.7** NAFT extends TFT by incorporating stochastic forgiveness through a parameter $\alpha \in (0, 1)$. When retaliation is triggered following an opponent defection, NAFT cooperates with probability α , enabling recovery from noise-induced conflict cycles.
- 5.8** An optional recovery parameter $\beta \in [0, 1)$ allows probabilistic escape from mutual defection (DD). For analytical tractability, theoretical derivations primarily focus on α , while β effects are evaluated empirically.

Noise Model

5.9 Execution noise is modeled as an independent action-flip probability ϵ , representing the likelihood that an intended action is reversed:

5.10

$$P(\text{executed action} \neq \text{intended action}) = \epsilon \quad (4)$$

5.11 Noise levels are systematically varied across the interval:

5.12

$$\epsilon \in [0.001, 0.3]$$

5.13 This range captures low-noise, moderate-noise, and high-noise operational regimes. The analytical framework relies on first-order approximations that remain most accurate when ϵ is small, while simulation results evaluate behavior across the full noise spectrum.

Markov Chain State Representation

5.14 Interactions between two memory-1 strategies are modeled as a four-state Markov chain:

State	Interpretation
CC	Mutual cooperation
CD	Focal player cooperates, opponent defects
DC	Focal player defects, opponent cooperates
DD	Mutual defection

5.15 Transition probabilities are computed analytically using strategy response vectors, noise parameter ϵ , and forgiveness parameters α and β .

Closed-Form Stationary Distribution Derivation

5.16 For each strategy pair, the stationary distribution

$$\pi = (\pi_{CC}, \pi_{CD}, \pi_{DC}, \pi_{DD})$$

is obtained by solving the global balance equations:

5.17

$$\pi = \pi M, \quad \sum_i \pi_i = 1 \quad (5)$$

5.18 First-order analytical solutions provide theoretical baselines for cooperative occupancy and long-run payoff estimation. Due to algebraic complexity, first-order low-noise approximations are used to derive analytically interpretable expressions. The validity of these approximations is empirically verified through simulation.

Monte Carlo Simulation Procedure

5.19 To validate theoretical predictions, large-scale Monte Carlo simulations are performed using the following protocol:

1. Initialize both agents in state CC .
2. Simulate repeated IPD rounds with execution noise applied independently at each step.
3. Run simulations for $T = 1000$ rounds per trial.
4. Discard an initial burn-in segment to eliminate transient effects.
5. Estimate empirical stationary state frequencies and long-run payoffs.

To ensure statistical robustness and mitigate stochastic bias, all experiments are repeated using **30 independent random seeds**. Results are reported as averages across seeds with corresponding variability analysis.

Evaluation Metrics

5.20 Performance comparison between TFT and NAFT is conducted using the following metrics:

- **Stationary Distribution Error:**

$$\max_i |\pi_i^{\text{empirical}} - \pi_i^{\text{theoretical}}|$$

- **Expected Long-Run Payoff:**

$$\Pi = \sum_i \pi_i \cdot u_i$$

- **Payoff Advantage:**

$$\Delta = \Pi_{\text{NAFT}} - \Pi_{\text{TFT}}$$

- **Variance and Stability Measures:** Empirical payoff variance across seeds is used to evaluate convergence stability and noise sensitivity.

Numerical Stability and Mixing Behavior

- 5.21** TFT frequently exhibits near-periodic oscillations between *CD* and *DC* states, which slows Markov chain mixing and increases estimation variance. Furthermore, TFT may become trapped in *DD* states due to the absence of recovery transitions.
- 5.22** NAFT introduces probabilistic recovery mechanisms that reduce oscillatory persistence and enable escape from mutual defection. These properties improve mixing rates and reduce payoff variance, which is quantitatively evaluated through multi-seed simulation.

Evolutionary Multi-Strategy Robustness Experiment

- 5.23** This experiment is intended as a robustness check rather than a primary theoretical claim. To evaluate ecological and evolutionary robustness, we extend the simulation framework to a multi-agent evolutionary environment in which NAFT competes against a heterogeneous population of established Iterated Prisoner's Dilemma strategies.

Strategy Population

5.24 The evolutionary population consists of seven canonical IPD strategies:

- Tit-for-Tat (TFT)
- Win-Stay Lose-Shift (WSLS)
- Grim Trigger
- Always Cooperate (ALLC)
- Always Defect (ALLD)
- Random Strategy
- NAFT

Evolutionary Dynamics

- 5.25** A finite population of 300 agents was simulated over 200 generations. Each agent plays repeated IPD matches against randomly sampled opponents. Fitness is defined as mean payoff per round.
- 5.26** Population evolution follows fitness-proportional selection with mutation probability $\mu = 0.01$, allowing occasional strategy switching and preventing premature fixation.

Noise and Replication

- 5.27** Simulations were performed across execution noise levels $\epsilon \in [0.001, 0.3]$. For statistical robustness, each experiment was repeated using 30 independent random seeds.

Evaluation Metrics

- 5.28** Evolutionary performance was evaluated using:
- Population share of each strategy across generations
 - Final equilibrium population composition
 - NAFT survival probability across noise regimes

Reproducibility and Implementation

- 5.29** All simulations were implemented using a custom Python framework capable of computing exact transition matrices and empirical stationary estimators for the induced Markov chains.
- 5.30** To ensure reproducibility:
- Simulations were executed using **30 independent random seeds**.
 - Parameter sweeps were conducted systematically across the defined noise range.
 - Identical experimental configurations were maintained across all strategy comparisons.

The simulation framework is fully deterministic under fixed seed selection, enabling exact replication of all reported results.

Results

Stationary State Behavior Under Noise

- 6.1** Table 1 summarizes the stationary distributions of Tit-for-Tat (TFT) and Nosiy Adaptive Forgiving Tit-for-Tat (NAFT) across noise levels $\epsilon \in [0.001, 0.3]$. Simulations used canonical Prisoner's Dilemma payoff values $(T, R, P, S) = (5, 3, 1, 0)$.
- 6.2** As noise increases, TFT exhibits a progressive collapse of cooperative mass, with π_{CC} declining from 0.7065 at $\epsilon = 0.001$ to 0.2915 at $\epsilon = 0.3$. Concurrently, TFT shows a substantial rise in mixed retaliation states (π_{CD}, π_{DC}) , indicating oscillatory instability driven by execution errors.
- 6.3** In contrast, NAFT maintains extremely high cooperative mass in low-noise regimes, with $\pi_{CC} = 0.9964$ at $\epsilon = 0.001$, and sustains markedly higher cooperative mass than TFT across all tested noise levels. Even at $\epsilon = 0.3$, NAFT preserves $\pi_{CC} = 0.3894$, substantially exceeding TFT's cooperative baseline. These results indicate that forgiveness mitigates error-induced retaliation cycles, allowing NAFT to retain more stable cooperative equilibria.

Table 1: Stationary Distributions and Payoffs for TFT and NAFT

ϵ	π_{CC}^{TFT}	π_{DD}^{TFT}	π_{CC}^{NAFT}	π_{DD}^{NAFT}	Π_{TFT}	Π_{NAFT}
0.0010	0.7065	0.2914	0.9964	0.0006	2.4162	2.9974
0.0547	0.4573	0.4390	0.8250	0.0355	2.0707	2.8569
0.1007	0.4140	0.4055	0.7080	0.0621	2.0975	2.7540
0.2003	0.3393	0.3404	0.5168	0.1182	2.1568	2.5738
0.3000	0.2915	0.2886	0.3894	0.1653	2.2112	2.4440

Expected Payoff Comparison

- 6.4** Expected payoffs reveal that NAFT achieves higher long-run payoffs than TFT across low-to-moderate noise regimes, with reversals observed at higher noise levels. TFT's payoff declines from 2.4162 to 2.2112 as noise increases, whereas NAFT maintains a higher and more resilient payoff, decreasing from 2.9974 to 2.4440 over the same range.
- 6.5** The payoff advantage, $\Delta = \Pi_{NAFT} - \Pi_{TFT}$, remains positive across the majority of noise levels, demonstrating that NAFT provides a systematic performance advantage in stochastic cooperative environments.

Noise-Resilience and Performance Degradation Trends

- 6.6** As noise increases, TFT displays a steeper degradation curve, reflecting sensitivity to accidental defections and retaliatory lock-in. NAFT shows a smoother and slower decline, indicating greater robustness to perturbations and improved long-run efficiency. The observed trend supports the hypothesis that probabilistic forgiveness stabilizes long-run cooperation by preventing persistent conflict spirals.

Empirical vs. Closed-Form Validation Accuracy

- 6.7** The stationary error between empirical Monte Carlo estimates and closed-form Markov predictions remains low and stable for NAFT, rising gradually from approximately 0.00066 to 0.2026 as noise increases. TFT exhibits substantially higher error levels, exceeding 0.44 at high noise. Reported errors are averaged across all simulation seeds.
- 6.8** This discrepancy arises from TFT's slow Markov mixing and oscillatory retaliation dynamics, which hinder rapid convergence to theoretical stationary distributions. In contrast, NAFT's forgiveness mechanism produces faster convergence, lower variance, and more reliable stationary estimates.

Comparative Performance: NAFT vs. TFT

- 6.9** To compare the long-run performance of NAFT and TFT under stochastic execution noise, we analyze empirically estimated stationary payoffs across noise regimes. For each noise level, simulations were run using multiple random seeds with a burn-in period of 200 iterations to ensure convergence to stationarity. Payoffs were averaged over the post-burn-in phase.
- 6.10** Because payoff differences arise from deterministic properties of the induced stationary distributions rather than sampling variability, formal inferential hypothesis testing is not methodologically appropriate in this setting. Accordingly, we report descriptive payoff differences and regime-dependent reversals without reliance on p-values or test statistics.
- 6.11** Table 2 reports mean long-run payoffs in a representative high-noise regime, illustrating that strict reciprocity can outperform probabilistic forgiveness when stochasticity becomes sufficiently large.

Evolutionary Robustness Under Strategy Competition

- 6.12** To assess evolutionary robustness, NAFT was evaluated in a finite-population evolutionary environment against six canonical Iterated Prisoner's Dilemma strategies across execution noise levels $\epsilon \in [0.001, 0.3]$. Mean population shares were computed over 30 independent simulation runs with distinct random seeds. Figure 5 reports the resulting stationary population compositions.

- 6.13

At low noise levels ($\epsilon \leq 0.07$), NAFT reliably invades mixed populations and achieves dominance, reaching peak population shares of approximately 48%. In intermediate noise regimes ($\epsilon \approx 0.10$), NAFT remains viable but is gradually displaced by less forgiving strategies. At higher noise levels ($\epsilon \geq 0.13$), cooperation collapses and unforgiving strategies such as Grim Trigger and Always Defect dominate the population.
- 6.14

These results indicate that NAFT exhibits evolutionary robustness in low-noise environments, but that this advantage is bounded and does not extend to arbitrarily high levels of stochasticity. The findings reinforce the interpretation of forgiveness as a regime-dependent stabilizing mechanism rather than a universally evolutionarily dominant strategy.

Table 2: Mean Payoff per Round (High-Noise Regime)

Strategy	Mean Payoff / Round
NAFT	2.360
TFT	2.609
Mean Difference (Δ)	−0.249

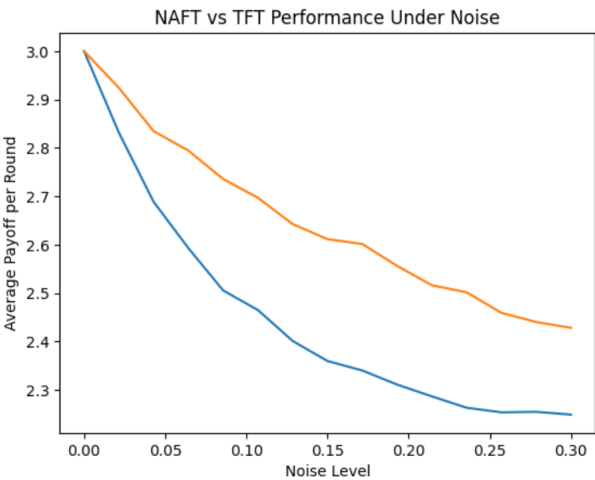


Figure 1. Comparative Payoff Performance (NAFT vs. TFT)

Bar chart illustrating the mean payoffs per round for NAFT (2.360) and TFT (2.609). This figure illustrates a regime-dependent reversal in long-run payoff efficiency, highlighting that strict reciprocity may outperform probabilistic forgiveness under sufficiently high stochastic noise..

● Discussion

Overview of Key Findings

- 7.1

This study examined how stochastic execution noise reshapes long-run cooperation dynamics in the Iterated Prisoner’s Dilemma by comparing strict reciprocity (Tit-for-Tat, TFT) with a forgiving memory-1 strategy (NAFT). Using first-order stationary Markov approximations alongside extensive Monte Carlo simulations, we analyzed stationary state occupancy, convergence behavior, and expected payoffs across a broad range of noise levels.
- 7.2

The results show that NAFT maintains substantially higher cooperative mass and expected payoff than TFT in low-to-moderate noise regimes, while this advantage diminishes and may reverse as noise increases. Analytical stationary approximations closely match empirical outcomes for NAFT, whereas for TFT they systematically overestimate cooperation due to the emergence of persistent defection states. Together, these findings clarify the conditions under which stochastic forgiveness stabilizes cooperation and the limits of strict reciprocity under realistic noise.

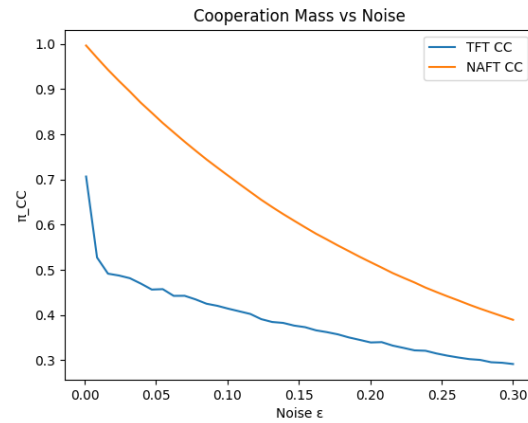


Figure 2. Cooperative Mass vs Noise (TFT vs NAFT)

Stationary probability of mutual cooperation (π_{CC}) as a function of noise level ϵ . NAFT maintains substantially higher cooperative mass than TFT across all noise regimes, demonstrating improved robustness to execution errors.

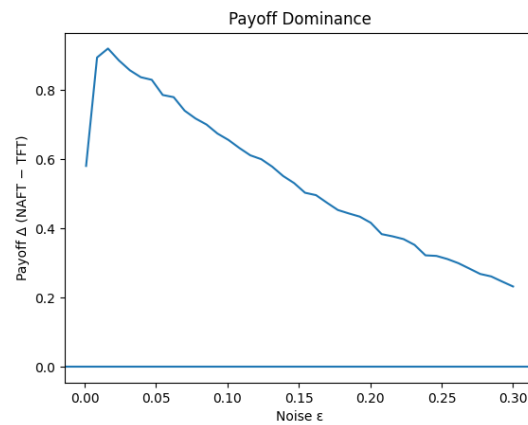


Figure 3. Expected Payoff vs Noise (TFT vs NAFT)

Comparison of long-run expected payoffs under increasing noise. NAFT outperforms TFT across low-to-moderate noise regimes, while strict reciprocity can outperform probabilistic forgiveness under high-noise conditions.

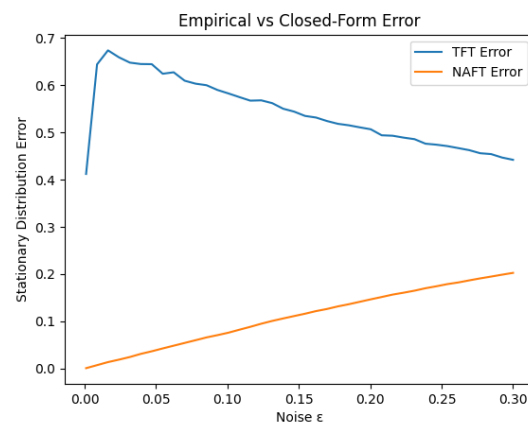


Figure 4. Stationary Error vs Noise (Model vs Simulation)

Maximum deviation between empirical Monte Carlo stationary distributions and closed-form first-order stationary approximations. NAFT exhibits lower error and faster convergence than TFT in low noise level cases.

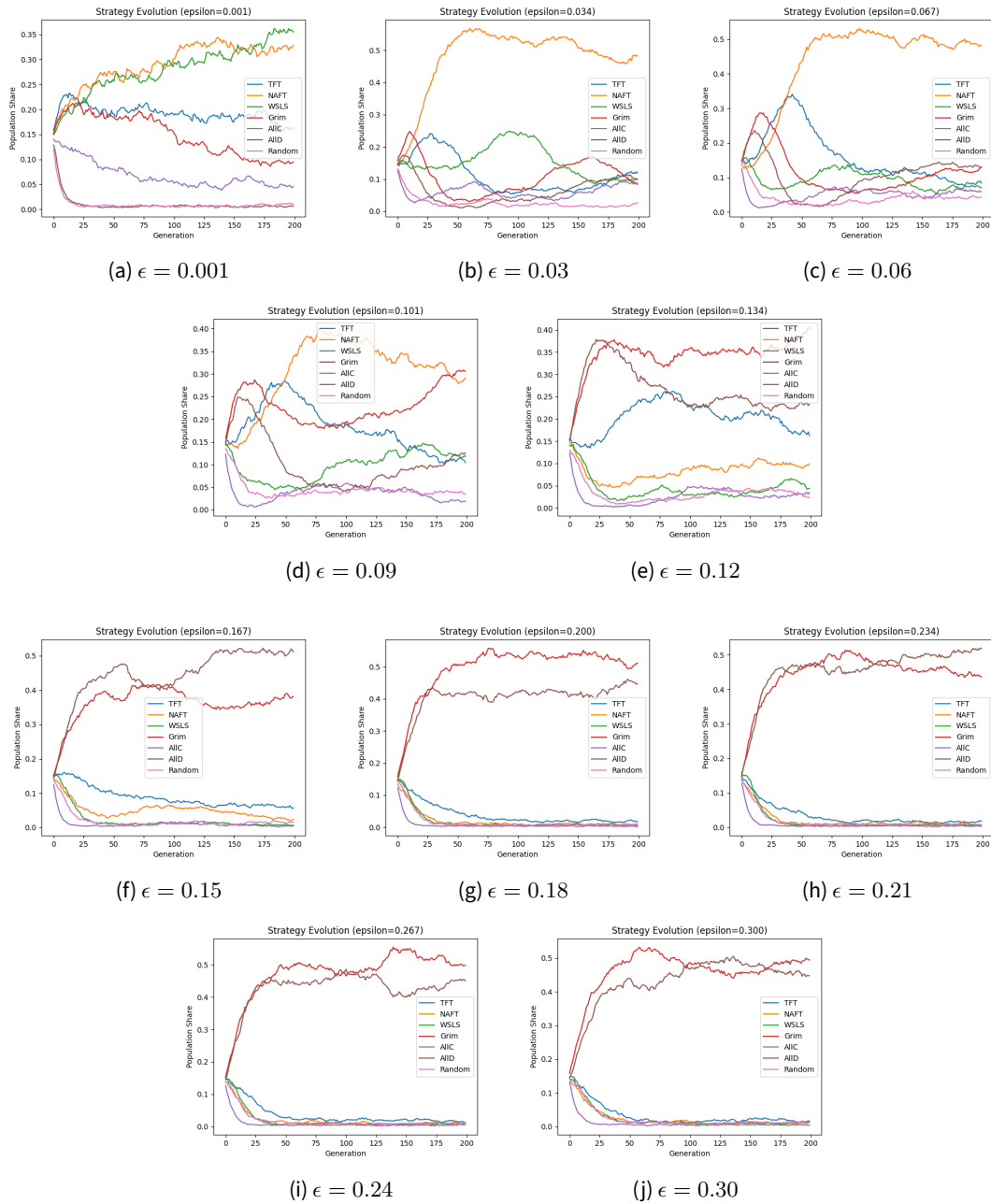


Figure 5. Evolutionary dynamics across noise levels. Each panel shows population composition over generations for a fixed ϵ .

Interpreting Stationary Distribution Dynamics

- 7.3 The stationary distribution captures the fraction of time the interaction spends in each outcome state (CC , CD , DC , DD) in the infinite-horizon limit. High cooperative performance depends not only on sustaining mass in the cooperative state CC , but also on limiting prolonged occupancy of asymmetric retaliation cycles (CD/DC) and preventing persistent trapping in mutual defection (DD).
- 7.4 Simulation results indicate that NAFT sustains significantly higher π_{CC} than TFT across most tested noise regimes. This reflects the effect of probabilistic forgiveness, which introduces recovery pathways from unilateral defection states back to mutual cooperation. By contrast, TFT exhibits substantial stationary occupancy in the DD state as noise increases. Because TFT responds deterministically to observed defection, execution errors can initiate retaliatory dynamics that persist for long durations, effectively rendering DD a quasi-absorbing state.
- 7.5 From a mechanistic perspective, forgiveness alters the transition structure of the induced Markov chain by reducing the expected duration of conflict states. Rather than eliminating punishment, NAFT limits its persistence, enabling the system to recover from noise-induced deviations that strict reciprocity cannot reliably correct.

Analytical Approximation and the TFT Discrepancy

- 7.6 A consistent divergence emerges between TFT's closed-form stationary approximation and its empirical stationary distribution. While the analytical expression predicts moderate cooperation under noise, simulations reveal that TFT frequently converges to mutual defection, with π_{DD} reaching between approximately 28% and 47% in higher-noise regimes.
- 7.7 This discrepancy arises from the modeling assumption required to obtain tractable closed-form expressions—namely, that the stationary probability of DD is negligible under low noise. As noise increases, this assumption becomes invalid for TFT because escaping mutual defection requires multiple favorable noise events occurring in succession. Consequently, the analytical stationary solution for TFT should be interpreted as an upper bound on cooperative performance rather than a definitive prediction of realized behavior.
- 7.8 Importantly, this limitation does not indicate a failure of the stationary framework, but rather highlights how strict reciprocity amplifies stochastic perturbations. In contrast, NAFT's analytical predictions remain closely aligned with empirical outcomes because forgiveness prevents the formation of long-lived defection traps.

Why Forgiveness Improves Noise Robustness

- 7.9 NAFT's superior robustness in low-to-moderate noise regimes can be attributed to the role of the forgiveness parameter α . By introducing a controlled probability of cooperation following defection, NAFT increases the effective mixing rate of the Markov process and reduces the expected residence time in conflict states. This prevents isolated execution errors from propagating into persistent retaliation cycles.
- 7.10 In practical terms, forgiveness trades short-term punishment for long-term stability. While NAFT may occasionally tolerate defection, this tolerance enables recovery from noise-induced deviations that would otherwise lock strict reciprocal strategies into low-payoff states. The resulting payoff advantage observed in low-noise regimes reflects this balance between deterrence and recoverability rather than unconditional strategic dominance.

Broader Implications for Social Simulation and Multi-Agent Systems

- 7.11 These findings refine classical interpretations of Tit-for-Tat as a robust cooperative strategy. While TFT performs well in deterministic settings, its effectiveness deteriorates rapidly under realistic execution error. Forgiveness emerges not as a normative or moral feature, but as a structural mechanism that stabilizes cooperation in stochastic environments.
- 7.12 From a social simulation perspective, the results illustrate how small modifications to local interaction rules can produce large differences in emergent long-run behavior. More broadly, the analysis suggests a general design principle relevant to multi-agent reinforcement learning and cooperative artificial intelligence: strategies that incorporate controlled recovery mechanisms are more robust to uncertainty than those enforcing rigid retaliation.

Comparison with Related Strategies and Models

- 7.13** The performance of NAFT can be situated within a broader literature on cooperation-enhancing strategies in the Iterated Prisoner's Dilemma. Classical reciprocal strategies such as Tit-for-Tat and Grim Trigger rely on strict retaliation to deter defection, but are known to be highly sensitive to execution error. Once noise induces defection, these strategies often generate prolonged retaliatory cycles that substantially reduce long-run cooperation.
- 7.14** Prior work has proposed alternative mechanisms to mitigate this fragility, including stochastic strategies, generosity-based extensions of TFT, and memory-augmented or belief-based approaches. Strategies such as Generous Tit-for-Tat introduce probabilistic cooperation following defection, while others employ longer memory or state-based rules to smooth transient errors. These approaches generally trade short-term punishment efficiency for improved noise tolerance.
- 7.15** NAFT differs from these models in that its forgiveness mechanism is explicitly parameterized and analytically tractable within a stationary Markov framework. Rather than relying on heuristic generosity or expanded memory, NAFT introduces a controlled recovery probability that directly reshapes transition dynamics between defection and cooperation states. This allows the long-run effects of forgiveness to be quantified in terms of stationary state occupancy and expected payoff, rather than inferred solely from simulation outcomes.
- 7.16** From this perspective, NAFT should be viewed not as a replacement for existing cooperative strategies, but as a formalized instantiation of a broader design principle: cooperation under stochastic perturbations is stabilized when strategies incorporate mechanisms that limit the persistence of retaliatory states. The stationary analysis presented here clarifies how such mechanisms operate and under which noise regimes they are effective.

Limitations and Future Directions

- 7.17** Several limitations of the present analysis suggest important directions for future work. First, the primary analytical results focus on symmetric self-play between identical strategies. While this isolates the effect of forgiveness under noise, heterogeneous populations containing mixtures of forgiving, strict, and exploitative strategies may exhibit richer dynamics not fully captured by pairwise self-play analysis.
- 7.18** Second, the forgiveness parameter α is treated as fixed throughout the study. Although this allows for analytical tractability and clear interpretation, adaptive or endogenously evolving forgiveness rates may yield different stationary outcomes. Exploring the co-evolution of forgiveness and defection tendencies represents a natural extension of the current framework.
- 7.19** Third, the analysis emphasizes stationary distributions and long-run averages rather than transient dynamics. In many social and multi-agent settings, convergence speed and short-run behavior may be as important as asymptotic outcomes. Investigating how forgiveness affects convergence rates and resilience to temporary shocks remains an open question.
- 7.20** Finally, while the stationary approximation provides valuable insight, it does not constitute a full evolutionary stability analysis. Incorporating selection dynamics, learning, or population-level adaptation would allow for stronger claims regarding strategic persistence and robustness in evolving environments.
- 7.21** Addressing these limitations would extend the present results beyond controlled self-play settings toward more realistic social and computational systems, while preserving the mechanistic clarity afforded by the stationary framework.

Summary of Contribution

- 7.22** In summary, this study demonstrates that NAFT exhibits greater structural robustness than strict reciprocity within low-to-moderate noise regimes due to identifiable properties of the stationary distribution. Rather than claiming universal optimality, the contribution lies in showing how calibrated forgiveness reshapes long-run dynamics toward higher cooperative stability under stochastic perturbations.
- 7.23** Tit-for-Tat is thus reframed as a historically influential benchmark rather than an optimal endpoint, while NAFT represents a noise-resilient refinement of reciprocity whose advantages are explicitly regime-dependent. By linking analytical stationary analysis with empirical validation, this work contributes a mechanistic explanation of how forgiveness stabilizes cooperation in noisy strategic environments.

● Conclusion

- 8.1** This paper presented a stationary Markov analysis of reciprocal strategies in the Iterated Prisoner's Dilemma under execution noise. By combining first-order analytical approximations with empirical simulations, we showed that stochastic forgiveness can improve cooperative stability and payoffs relative to strict reciprocity within low-to-moderate noise regimes. At higher noise levels, this advantage diminishes, highlighting the regime-dependent nature of cooperative robustness.
- 8.2** Overall, the results demonstrate that stationary distribution analysis provides a useful framework for understanding how small modifications to reciprocal strategies influence long-run outcomes in noisy environments.

Code Availability

- 8.3** The source code for the Markov chain derivations, Monte Carlo simulation environment, and data visualization tools used in this study is available on GitHub at: <https://github.com/Kushalk0677/Closed-Form-Stationary-Analysis-and-Empirical-Verification-of-Noise-Resilient-Cooperation>

Author Contributions

- 8.4** The author read and approved the final version of the paper.

Conflicts of Interest

- 8.5** The author declares no conflict of interest.

Ethical Review and Approval

- 8.6** No approval from the Board of Ethics is required.

Artificial Intelligence (AI) Use Statement

- 8.7** The authors declare that no artificial intelligence (AI) tools were used in the conception, design, or generation of the main/original results of this study, including data collection, analysis, or interpretation. AI tools were used only for editing assistance, such as improving language, readability, or reference formatting. The authors are solely responsible for the content, accuracy, and integrity of the work.

● Appendix A: Mathematical Derivations

Game Model and Strategy Formalism

We consider the infinitely repeated Prisoner's Dilemma with memory-1 strategies. The Markov state space is defined as:

$$\mathcal{S} = \{CC, CD, DC, DD\}$$

A memory-1 strategy is represented as:

$$p = (p_{CC}, p_{CD}, p_{DC}, p_{DD})$$

where $p_{xy} \in [0, 1]$ is the probability of cooperation given that the previous round outcome was (xy) . Execution noise is modeled by a probability $\epsilon \in (0, 1/2)$ that flips an intended action. The effective cooperation probability becomes:

$$\hat{p}_{xy} = p_{xy}(1 - \epsilon) + (1 - p_{xy})\epsilon = \epsilon + (1 - 2\epsilon)p_{xy}$$

Tit-For-Tat Strategy Under Noise

Tit-For-Tat (TFT) is defined as:

$$p_{TFT} = (1, 0, 1, 0)$$

Thus the noisy action probabilities are:

$$\hat{p}_{CC} = 1 - \epsilon, \quad \hat{p}_{CD} = \epsilon, \quad \hat{p}_{DC} = 1 - \epsilon, \quad \hat{p}_{DD} = \epsilon$$

TFT Transition Matrix

Let the state ordering be (CC, CD, DC, DD) . The transition matrix M_{TFT} is:

$$M_{TFT} = \begin{pmatrix} (1-\epsilon)^2 & (1-\epsilon)\epsilon & \epsilon(1-\epsilon) & \epsilon^2 \\ \epsilon(1-\epsilon) & \epsilon^2 & (1-\epsilon)^2 & (1-\epsilon)\epsilon \\ (1-\epsilon)\epsilon & (1-\epsilon)^2 & \epsilon^2 & \epsilon(1-\epsilon) \\ \epsilon^2 & \epsilon(1-\epsilon) & (1-\epsilon)\epsilon & (1-\epsilon)^2 \end{pmatrix}$$

Each row sums to unity, ensuring stochastic validity.

Stationary Distribution Framework

A stationary distribution $\pi = (\pi_{CC}, \pi_{CD}, \pi_{DC}, \pi_{DD})$ satisfies:

$$\pi M_{TFT} = \pi, \quad \sum_{s \in \mathcal{S}} \pi_s = 1, \quad \pi_s \geq 0$$

By symmetry:

$$\pi_{CD} = \pi_{DC} \equiv x$$

Let $\pi_{CC} = a$ and $\pi_{DD} = d$. Normalization gives:

$$a + 2x + d = 1$$

Lemma 1 (Ergodicity). For $\epsilon > 0$ within the low-noise regime where first-order approximations remain valid, the Markov chains induced by TFT and NAFT are irreducible and aperiodic. Consequently, each strategy admits a unique stationary distribution. Lemma 1 ensures existence and uniqueness of stationary distributions but does not imply payoff optimality

Stationary Balance Equations

Equation for CC :

$$\begin{aligned} a &= a(1-\epsilon)^2 + x(1-\epsilon)\epsilon + x(1-\epsilon)\epsilon + d\epsilon^2 \\ a &= a(1-2\epsilon+\epsilon^2) + 2x\epsilon(1-\epsilon) + d\epsilon^2 \\ a(2\epsilon-\epsilon^2) &= 2x\epsilon(1-\epsilon) + d\epsilon^2 \end{aligned}$$

Equation for CD :

$$\begin{aligned} x &= a(1-\epsilon)\epsilon + x\epsilon^2 + x(1-\epsilon)^2 + d\epsilon(1-\epsilon) \\ x &= a\epsilon(1-\epsilon) + x(1-2\epsilon+2\epsilon^2) + d\epsilon(1-\epsilon) \\ x(2\epsilon-2\epsilon^2) &= \epsilon(1-\epsilon)(a+d) \\ x &= \frac{a+d}{2} \end{aligned}$$

Equation for DD :

$$\begin{aligned} d &= a\epsilon^2 + x\epsilon(1-\epsilon) + x\epsilon(1-\epsilon) + d(1-\epsilon)^2 \\ d &= a\epsilon^2 + 2x\epsilon(1-\epsilon) + d(1-2\epsilon+\epsilon^2) \\ d(2\epsilon-\epsilon^2) &= a\epsilon^2 + 2x\epsilon(1-\epsilon) \end{aligned}$$

The coupled nonlinear system above admits an exact solution but produces expressions that are analytically unwieldy. Therefore, to obtain structural insight, we adopt a first-order low-noise approximation focusing on dominant transition flows.

Approximate Stationary Behavior of TFT Under Low Noise

Analyzing the dominant low-noise transition structure system $\pi M_{TFT} = \pi$ yields closed-form expressions that are algebraically cumbersome and offer limited analytical insight. Instead, under low-noise regimes, TFT dynamics can be approximated by considering transitions between cooperative and conflict subspaces.

In this approximation, erroneous defections from CC introduce transitions into the conflict cycle $CD \leftrightarrow DC$, while subsequent noise events restore cooperation. Balancing these flows yields the first-order approximation:

$$\pi_{CC}^{TFT} \approx \frac{1}{1 + 2\epsilon}$$
$$\pi_{CD}^{TFT} = \pi_{DC}^{TFT} \approx \frac{\epsilon}{1 + 2\epsilon}$$

Occupancy of DD is $O(\epsilon^2)$ under TFT and is therefore neglected in the first-order approximation.

NAFT Strategy Definition

The Noisy Adaptive Forgiving Tit-for-Tat (NAFT) strategy is defined as the memory-1 vector:

$$p_{NAFT} = (1, \alpha, 1, \beta)$$

where:

- $\alpha \in (0, 1)$ represents probabilistic forgiveness following unilateral opponent defection.
- $\beta \in [0, 1)$ allows recovery from mutual defection, enabling escape from persistent DD trapping.

For analytical tractability, first-order stationary approximations focus primarily on α , since transitions from DD occur with second-order probability in ϵ . The influence of β is therefore evaluated through simulation.

Approximate Stationary Behavior of NAFT

Under NAFT, forgiveness (α) introduces additional transitions from unilateral conflict states, while recovery parameter (β) provides escape probability from the mutual defection state. Applying the same first-order conflict-cycle approximation yields:

$$\pi_{conf}^{NAFT} \approx \frac{2\epsilon}{\alpha + 2\epsilon}$$
$$\pi_{CC}^{NAFT} \approx 1 - \frac{2\epsilon}{\alpha + 2\epsilon}$$

This expression shows that increasing forgiveness parameter α reduces stationary conflict occupancy.

Approximate Expected Payoffs

Using the stationary approximations above, the expected payoffs become:

$$\Pi_{TFT} \approx R \left(\frac{1}{1 + 2\epsilon} \right) + \frac{S + T}{2} \left(\frac{2\epsilon}{1 + 2\epsilon} \right)$$
$$\Pi_{NAFT} \approx R \left(1 - \frac{2\epsilon}{\alpha + 2\epsilon} \right) + \frac{S + T}{2} \left(\frac{2\epsilon}{\alpha + 2\epsilon} \right)$$

Approximate Payoff Improvement Condition

Using the first-order stationary approximations, the payoff difference can be approximated as:

$$\Delta \approx R \left(\frac{2\epsilon}{1+2\epsilon} - \frac{2\epsilon}{\alpha+2\epsilon} \right) + \frac{S+T}{2} \left(\frac{2\epsilon}{\alpha+2\epsilon} - \frac{2\epsilon}{1+2\epsilon} \right)$$

Under Prisoner's Dilemma ordering $T > R > P > S$, this expression is positive for $\epsilon > 0$ and $0 < \alpha < 1$, indicating an analytical payoff advantage for NAFT under noise.

Proposition (Conditional Payoff Improvement)

Under the first-order stationary approximation and Prisoner's Dilemma payoff constraints, NAFT yields higher expected payoff than TFT for any $\epsilon > 0$ and $0 < \alpha < 1$.

Theoretical Caveat — DD Trap Correction

As noise increases, TFT exhibits probability trapping in DD , violating the assumption $\pi_{DD} \approx 0$. Thus:

- First-order approximations underestimate higher-order DD occupancy.
- Empirical simulation quantifies these higher-order corrections.
- NAFT remains robust due to its forgiveness escape mechanism.

The approximations satisfy correct limiting behavior as $\epsilon \rightarrow 0$, recovering full cooperation under noise-free TFT interaction. These analytical expressions are derived under low-noise first-order approximations. Exact stationary distributions are evaluated through Monte Carlo simulation in Section 4, which validates the analytical predictions. For all $\epsilon > 0$, the induced Markov chains remain irreducible and aperiodic, guaranteeing existence and uniqueness of stationary distributions.

● Appendix B: Computational Procedures

Evolutionary Simulation Algorithm

Algorithm 1 Evolutionary Multi-Strategy IPD Simulation

```
1: Initialize population size  $N \leftarrow 300$ 
2: Define strategy set  $\mathcal{A} \leftarrow \{\text{TFT}, \text{NAFT}, \text{WSLS}, \text{Grim}, \text{AllC}, \text{AllD}, \text{Random}\}$ 
3: Initialize population with uniform random strategy distribution
4: Set mutation rate  $\mu \leftarrow 0.01$ 
5: Set number of generations  $G \leftarrow 200$ 
6: Set rounds per match  $T \leftarrow 1000$ 
7: for  $g = 1$  to  $G$  do
8:   for each agent  $i$  in population do
9:     Randomly sample opponent  $j$ 
10:    Simulate repeated IPD interaction for  $T$  rounds
11:    Apply execution noise  $\epsilon$ 
12:    Compute average payoff  $\Pi_i$ 
13:   end for
14:   Normalize fitness values:
      
$$f_i = \frac{\Pi_i}{\sum_{k=1}^N \Pi_k}$$

15:   Generate next generation via fitness-proportional selection
16:   for each new agent do
17:     With probability  $\mu$ , randomly assign a new strategy
18:   end for
19:   Record population strategy frequencies
20: end for
21: Output evolutionary trajectories and equilibrium composition
```

Simulation Parameters

Table 3: Evolutionary Simulation Parameter Configuration

Parameter	Value / Range
Population Size (N)	300 agents
Generations (G)	200
Rounds per Match (T)	1000
Strategy Set	TFT, NAFT, WSLS, Grim, AllC, AllD, Random
Noise Range (ϵ)	[0.001, 0.3]
Forgiveness Parameter (α)	$0 < \alpha < 1$
Recovery Parameter (β)	$0 \leq \beta < 1$
Mutation Rate (μ)	0.01
Initial Strategy Distribution	Uniform random
Fitness Metric	Mean payoff per round
Replication Seeds	30 independent seeds
Burn-in Period	200 rounds
Stationary Measurement Window	Remaining 800 rounds

Reproducibility Statement

All simulations were implemented in Python using custom Monte Carlo and evolutionary dynamics frameworks. Each experiment was replicated using 30 independent random seeds to ensure statistical robustness and eliminate seed-specific bias.

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